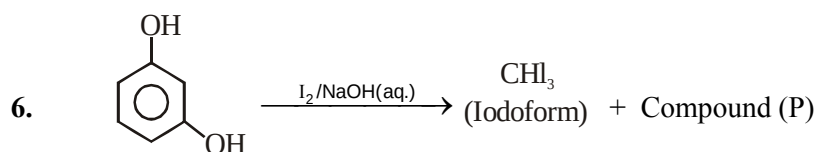
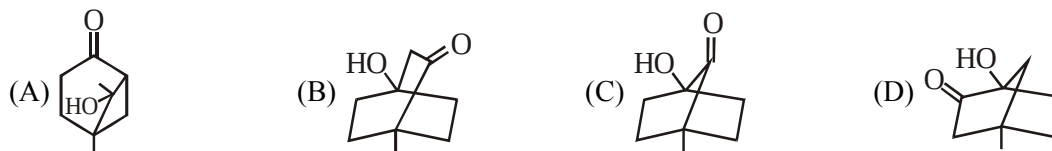
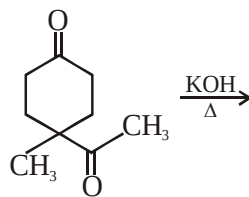
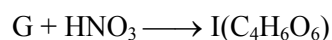
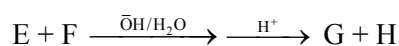
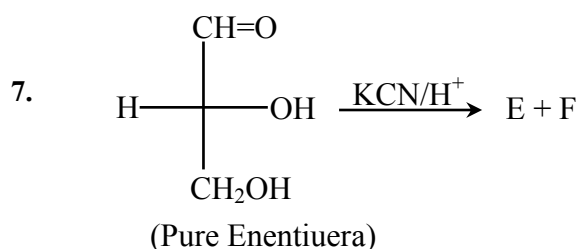
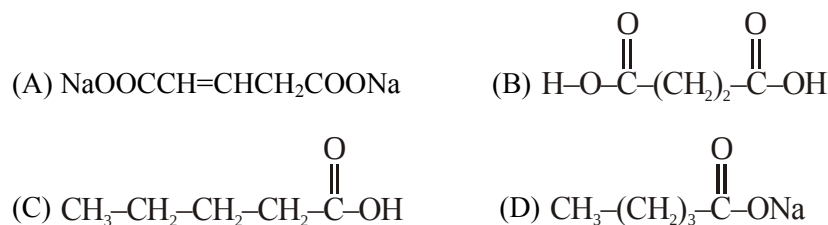


5. The major product of the following reaction is –



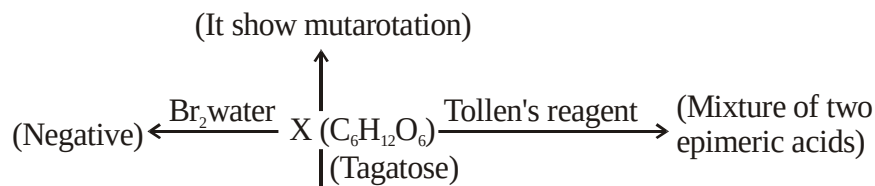
P is :



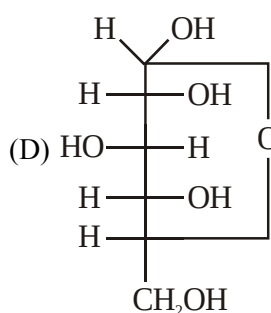
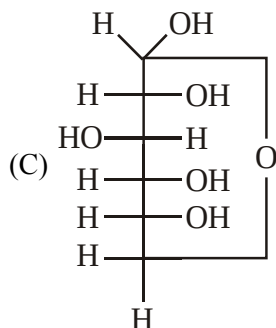
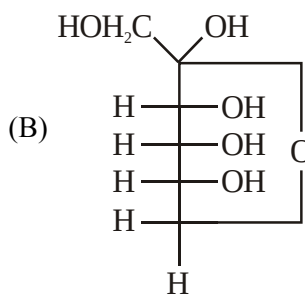
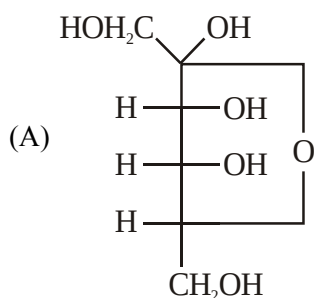
Molecular formula of both E and F is $C_4H_7NO_3$ and are chiral which of the following is/are correct for the above reactions ?

- (A) E and F are enantiomers of each other.
- (B) G and H have different melting point.
- (C) I is achiral.
- (D) G is erythro isomer and H is threo isomer.

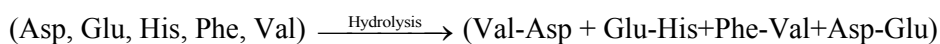
8. Observe the following road map and identify the carbohydrate 'X' (Tagatose)



(has pyranose structure)



9. A pentapeptide contains five type of α -amino acids. It gives following dipeptides only on hydrolysis.



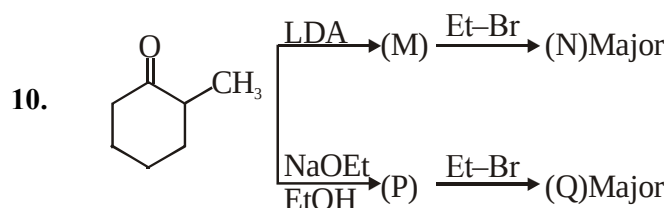
The correct sequence of amino acids in the polypeptide is :

(A) Asp-Glu-His-Phe-Val

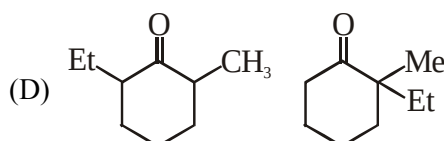
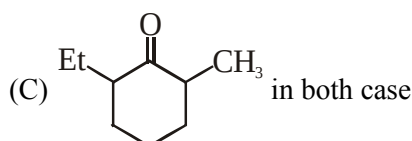
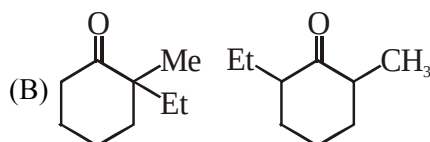
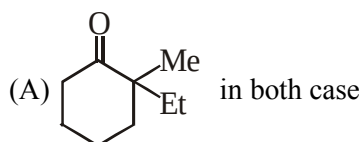
(B) Phe-Val-Asp-Glu-His

(C) Val-Asp-Glu-His-Phe

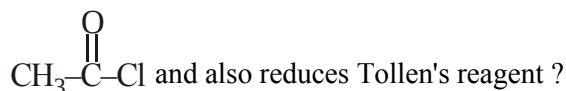
(D) Glu-His-Phe-Val-Asp



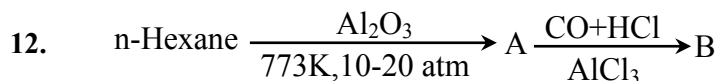
(N) and (Q) are respectively :



11. In which of the following carbohydrates, molecular weight is increased by 210, when react with excess of



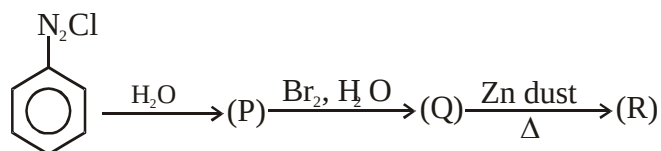
- (A) α -D-Glucose (B) D-Fructose (C) β -D-Fructose (D) α -D-Maltose



Select the correct statement among following ?

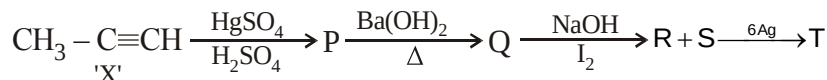
- (A) Compound 'B' form silver mirror on reaction with $[\text{Ag}(\text{NH}_3)_2]\text{OH}$.
 (B) Molecularity of reaction during conversion from 'A' to 'B' is '3'.
 (C) Compound 'A' can also be synthesised by reaction of benzene diazonium chloride with H_3PO_2 .
 (D) Compound 'B' give grey colour with aq. HgCl_2 solution.

13. The dipole moment of products obtained in the given reaction is -

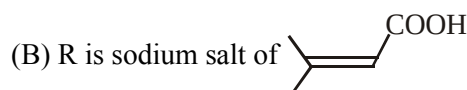


- (A) $\mu_{\text{R}} \neq 0$ (B) $\mu_{\text{R}} = 0$ (C) $\mu_{\text{R}} < \mu_{\text{Q}}$ (D) $\mu_{\text{R}} > \mu_{\text{Q}}$

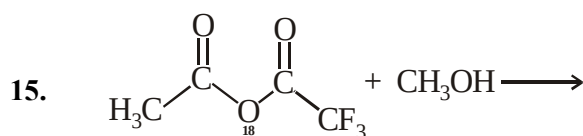
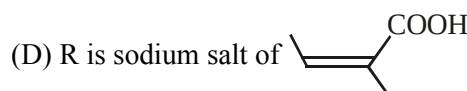
14. Correct option regarding following reaction sequence.



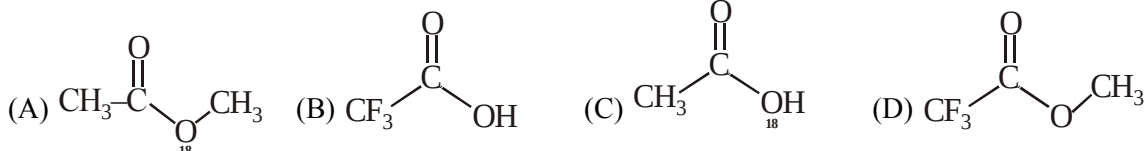
- (A) T is homologue of 'X'



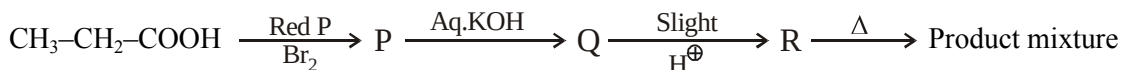
- (C) P can produce S with NaOI



Product is/are :

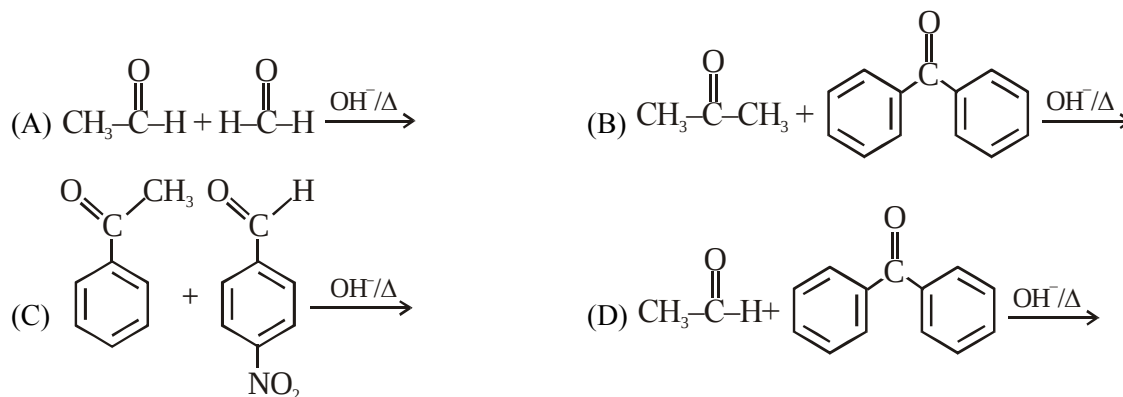


16. In following reaction final product mixture contains -

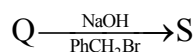


- (A) Two optically active organic compounds. (B) One optically inactive organic compound.
 (C) Two optically inactive organic compounds. (D) Three stereoisomers are formed.

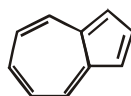
17. Which of the following will give crossed aldol product as major product ?



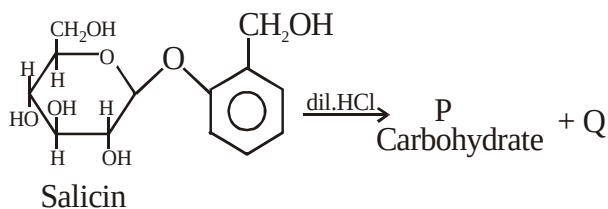
18. The correct statement(s) about the following reaction sequence is(are) -



- (A) R is steam volatile.
- (B) Q gives dark violet coloration with 1% aqueous FeCl_3 solution.
- (C) S gives yellow precipitate with 2, 4-dinitrophenylhydrazine.
- (D) S gives dark violet coloration with 1% aqueous FeCl_3 solution.
19. Which of the following statement/s is/are correct regarding Azulene ?

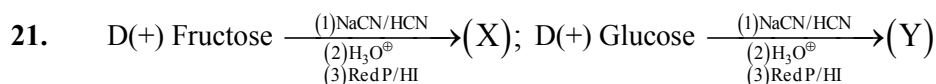


- (A) Azulene is an isomer of naphthalene and is blue in colour.
- (B) Azulene has very high dipole moment while naphthalene has dipole moment zero.
- (C) Seven membered ring of azulene is electrophilic and five membered ring is nucleophilic.
- (D) Seven membered ring of azulene is nucleophilic and five membered ring is electrophilic.
20. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain. Observe the following reaction of salicin.



The correct statement (s) is (are)

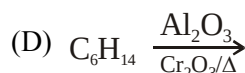
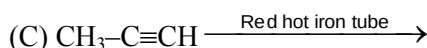
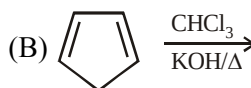
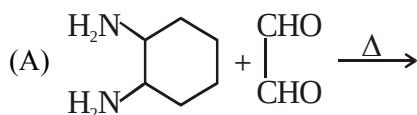
- (A) P is D- glucose.
- (B) Q is 2-hydroxybenzylalcohol.
- (C) Q is 2-Methoxy benzylalcohol.
- (D) The above reaction occurs through a carbocation.



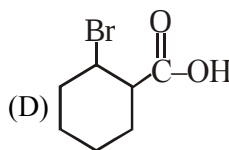
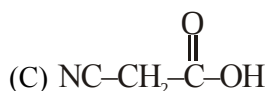
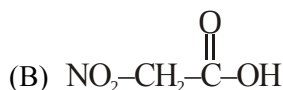
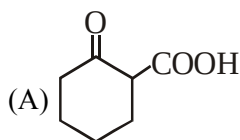
Select the **incorrect** statement regarding products X and Y ?

- (A) Both are position isomers (B) Both are conformational isomers
(C) Both are diastereomers (D) Both are chain isomers

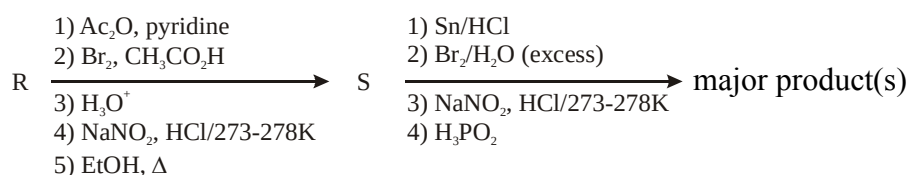
22. Which of the following may produce aromatic compound?



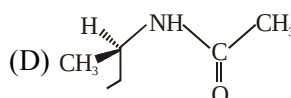
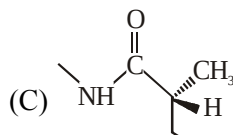
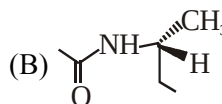
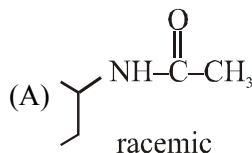
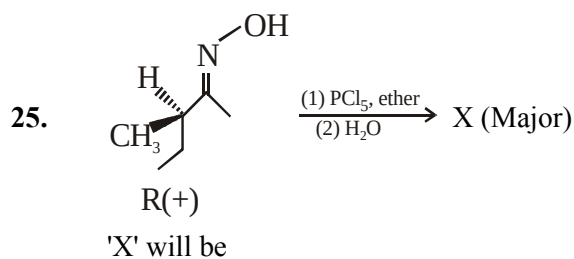
23. Which of the following will decarboxylate easily on heating only or heating in presence of OH^- ?



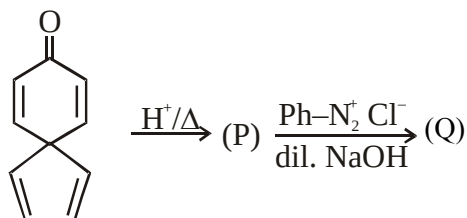
24. Aniline reacts with mixed acid (conc. HNO_3 and conc. H_2SO_4) at 288 K to give three isomeric products (P, Q & R). Product R is the major product. Consider the following reactions and choose the correct options.



- (A) Product (S) is more reactive than product (T) towards nucleophilic reactions.
(B) Molecular formula of product (S) is $C_6H_4NO_2Br$.
(C) Molecular formula of product (T) is $C_6H_3Br_3$.
(D) Meta product of aniline (in initial step) is obtained in significant amount as compare to ortho product.



26. The correct statement(s) about the following reaction is/are :



(A) P on reaction with Zn dust gives Naphthalene.

(B) Q is red colour dye.

(C) P is β -Naphthol.

(D) Q on reaction with ethyl alcohol can give (P).

27. Treatment of benzene with CO/HCl in the presence of anhydrous $\text{AlCl}_3/\text{CuCl}$ followed by reaction with $\text{Ac}_2\text{O}/\text{NaOAc}$ gives compound X as the major product. Compound X upon reaction with $\text{Br}_2/\text{Na}_2\text{CO}_3$, followed by heating at 473 K with moist KOH furnishes Y as the major product. Reaction of X with $\text{H}_2/\text{Pd-C}$, followed by H_3PO_4 treatment gives Z as the major product. Choose the correct options about the above reactions.

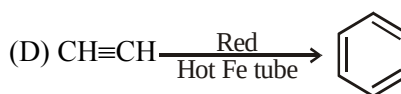
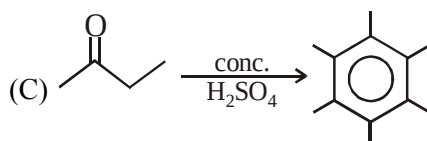
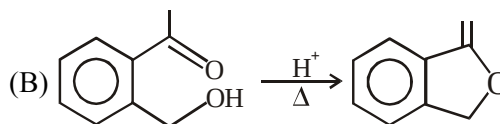
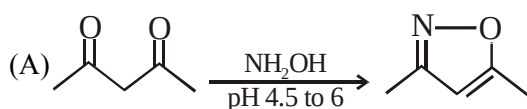
(A) Compound X shows stereoisomerism.

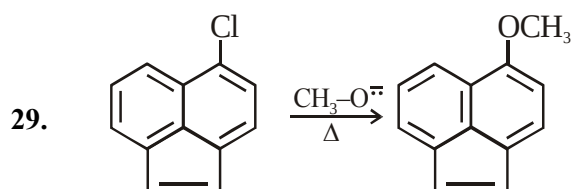
(B) Compound Y is a hydrocarbon.

(C) Compound Z gives iodoform with NaOI.

(D) Compound Y gives white precipitate with Tollen's reagent.

28. Identify reaction correctly match with its product ?





Which of the following statement/s is/are correct about above reaction?

(A) This reaction is not possible as no electron withdrawing group is present at ortho or para to leaving group $-\text{Cl}$.

(B) The nucleophile CH_3O^- is not strong enough to deprotonate the aromatic ring so benzyne mechanism is ruled out.

(C) $\text{S}_{\text{N}}1$ like reaction is not possible because $-\text{Cl}$ here is not good leaving group due to double bond character between carbon and chlorine.

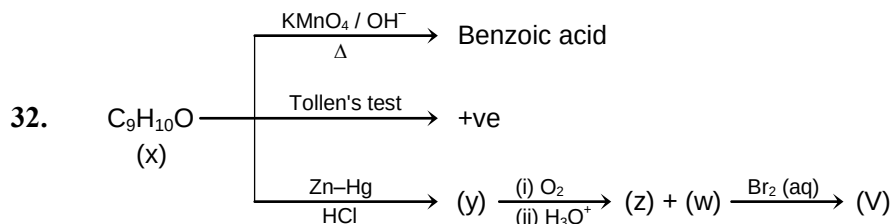
(D) This reaction takes place by $\text{ArS}_{\text{N}}2$ mechanism as meisenheimer complex renders five membered ring aromatic as in cyclopentadiene ion.

30. In which of the following polymers the strong intermolecular hydrogen bonding is present ?

(A) Cellulose (B) Nylon (C) PVC (D) Polyisoprene

31. Rhamnose ($\text{MM} = 164$) $\xrightarrow[\text{excess}]{(\text{CH}_3\text{CO})_2\text{O}}$ Acetylated product ($\text{MM} = 332$)

Find the number of hydroxy groups in a molecule of Rhamnose.

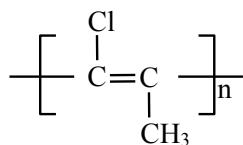


Compound (z) is smallest ketone and compound (x) has one methyl group as a substituent.

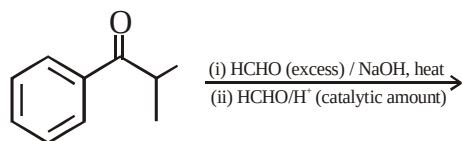
What is the maximum number of moles of Br_2 , Which can react with 1 mole of (W)

33. Molecular mass of the monomers used in the following polymer is

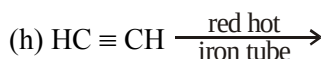
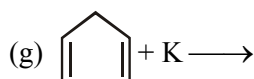
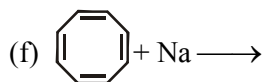
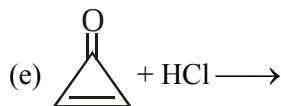
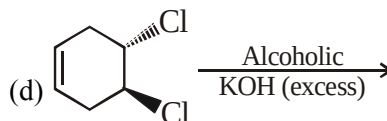
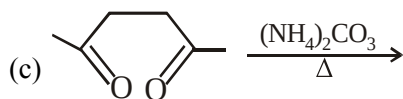
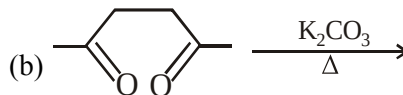
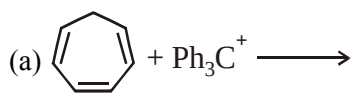
[Consider atomic mass of $\text{C} = 12$, $\text{H} = 1$, $\text{Cl} = 35.5$]



34. Molecular mass of the major product of the following reaction sequence is -

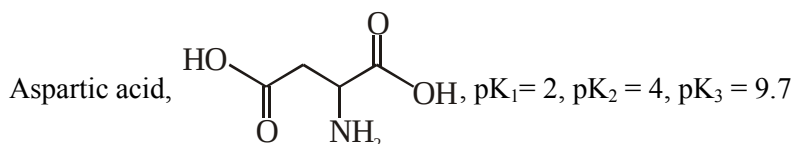


35. How many of the following reactions give aromatic product ?

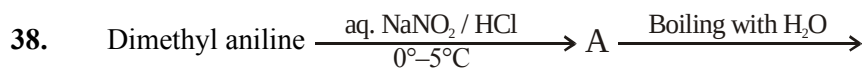


36. Identify total number of ester with $\text{C}_5\text{H}_{10}\text{O}_2$ molecular formula which give 3° alcohol as product on reaction with with excess of MeMgCl followed by H_2O ?

37. Consider the following amino acid :



Isoelectric point of Aspartic acid is-



How many total number of products are obtained after the completion of reaction ?

39. How many of the following compounds will not show mutarotation ?

(i) $\alpha\text{-D}(-)$ Glucopyranose (ii) $\beta\text{-D}(-)$ Glucopyranose (iii) $\alpha\text{-D}(-)$ Fructofuranose

(iv) $\beta\text{-D}(+)$ Fructofuranose (v) $\beta\text{-D}(-)$ Galactopyranose

(vi) Methyl $\alpha\text{-D}$ -Glucopyranoside

(vii) Sucrose

(viii) Maltose

(ix) Starch

40. How many of the following are condensation co-polymer ?

(i) Terylene

(ii) Nylon-66

(iii) Buna-N

(iv) Novalac

(v) Natural rubber

(vi) Bakelite

(vii) Buna-S

(viii) Glyptal

(ix) Melamine

41. How many of the following amino acid migrates towards cathode at $\text{pH} = 6$ on electrophoresis ?

(i) Glycine

(ii) Isoleucine

(iii) Arginine

(iv) Lysine

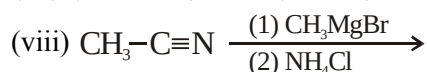
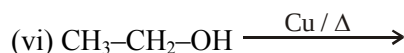
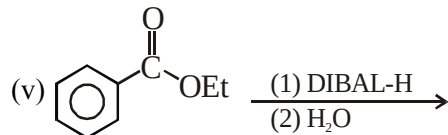
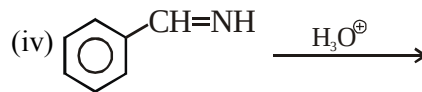
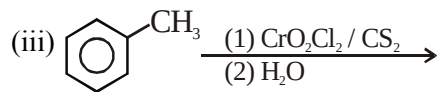
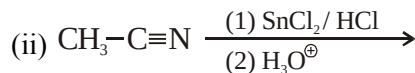
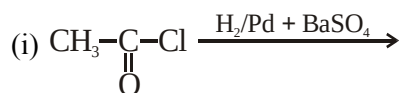
(v) Aspartic acid

(vi) Alanine

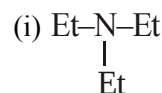
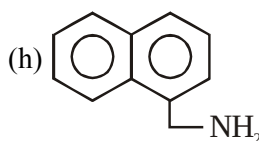
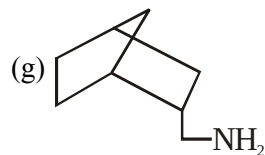
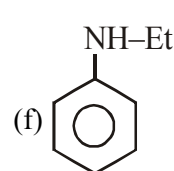
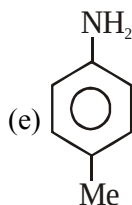
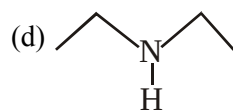
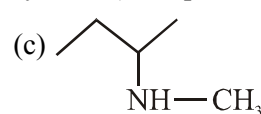
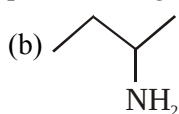
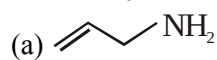
(vii) Glutamic acid

(viii) Histidine

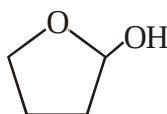
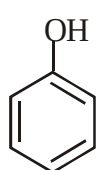
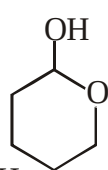
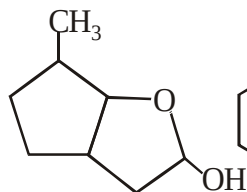
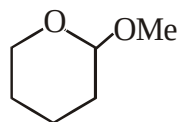
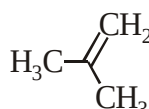
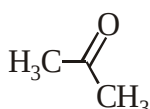
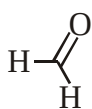
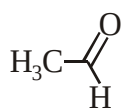
42. In how many reactions aldehyde is obtained as one of the products ?



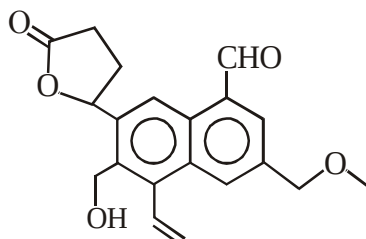
43. How many of the following compounds will give isocyanide (carbylamine) test positive ?



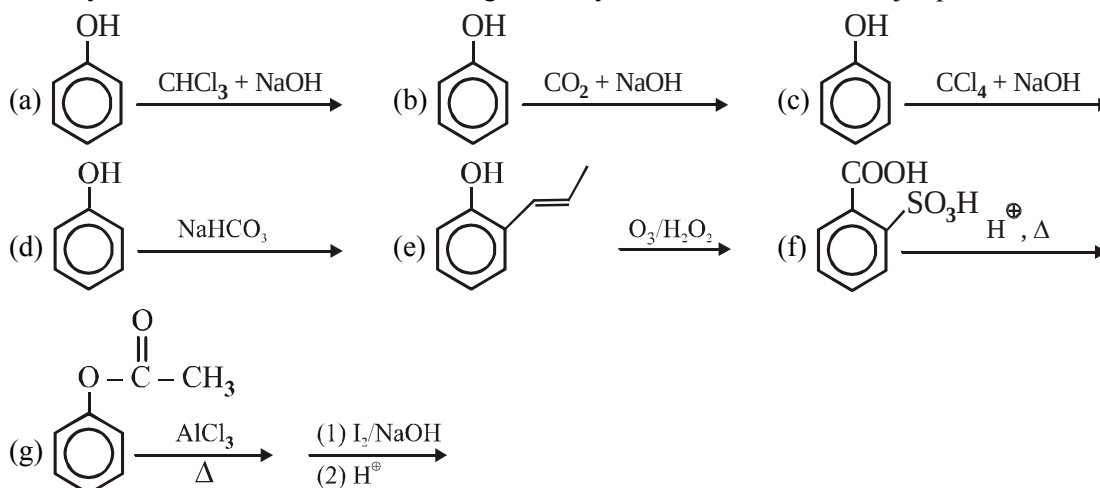
44. Identify total number of compounds which give positive test with Tollens reagent (silver mirror test) ?



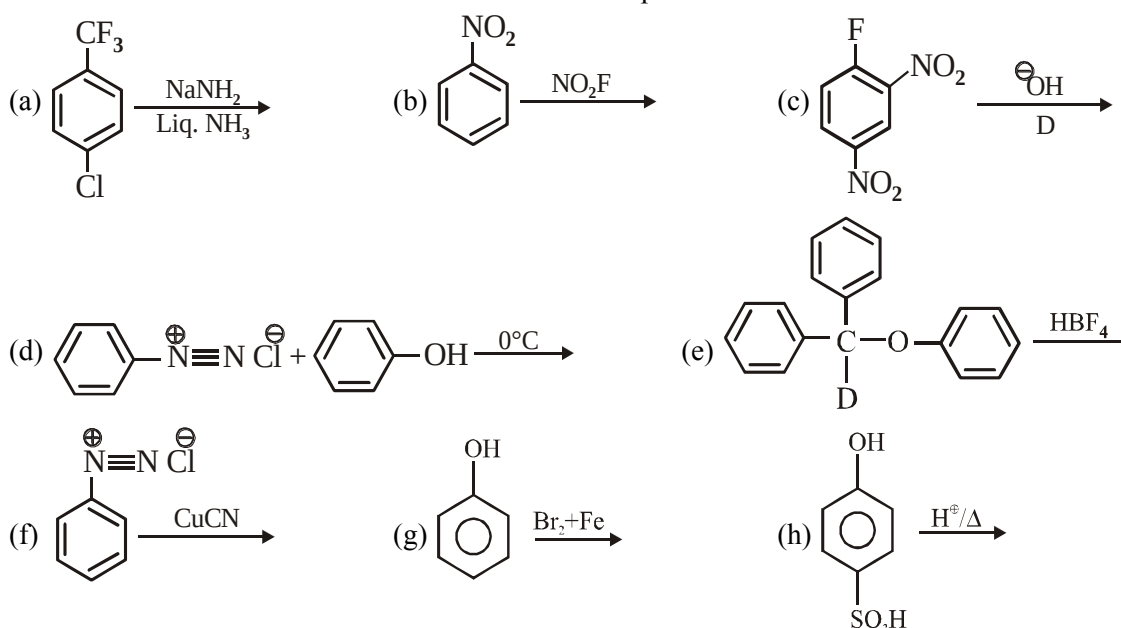
45. Total number of chlorine atoms present in all organic products obtained when given compound is reacted with excess PCl_5 .



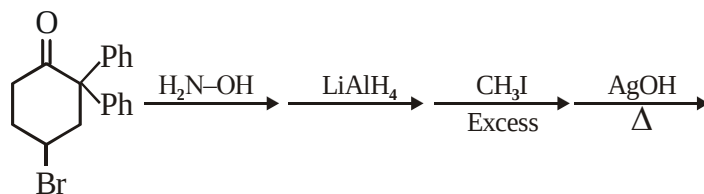
46. Identify number of reactions that would give salicylic acid or its salts as major product ?



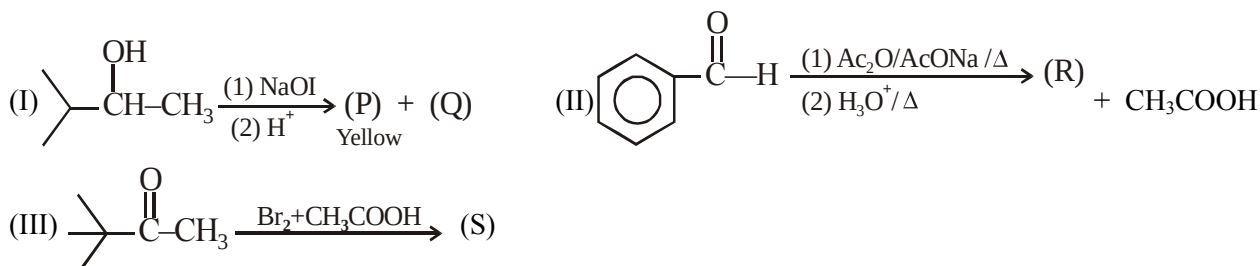
47. Find out number of reactions that are aromatic electrophilic substitution in nature ?



48. How many hyperconjugable hydrogen atoms are present in final product ?



49. Observe the following reactions ?

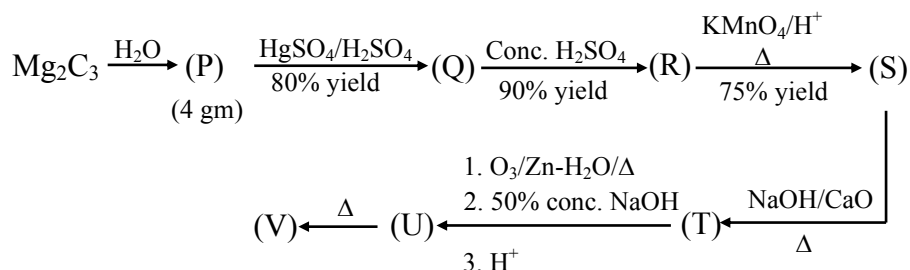


Degree of unsaturation in compounds (Q + R + S) is ?

50. Phenol react with phthalic anhydride in presence of H_2SO_4 and forms a compound (P). How many sp^2 hybridised carbon atoms are present in compound (P) ?

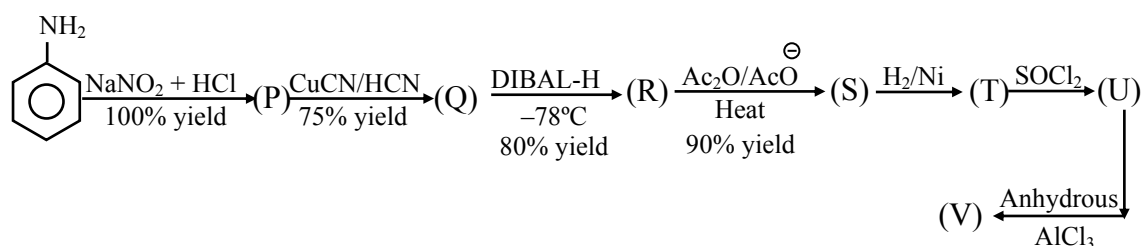
Stem Question

Stem Question (Q.51 to Q.52)



51. When 4 gm of (P) is taken then mass of (S) in grams is-
52. What is degree of unsaturation of compound (V) ?

Stem Question (Q.53 to Q.54)



53. When 93 gm of aniline is taken then mass of (S) in gm is
54. Number of sp^2 hybrid carbon atoms in compound (V) is

Answer Key

1. (B)	2. (D)	3. (C)	4. (B)	5. (B)
6. (A)	7. (B)	8. (B)	9. (B)	10. (D)
11. (A,B,C)	12. (A,C,D)	13. (B,C)	14. (A,B,C)	15. (C,D)
16. (A,B,D)	17. (A,C)	18. (B,C)	19. (A,B,C)	20. (A,B,D)
21. (A,B,C)	22. (B,C,D)	23. (A,B,C,D)	24. (A,B,D)	25. (B,D)
26. (A,B,C)	27. (A,B,D)	28. (A,B,C,D)	29. (B,C,D)	30. (A,B)
31. 4	32. 3	33. 74.5	34. 208	35. 7
36. 6	37. 3	38. 6	39. 3	40. 6
41. 3	42. 7	43. 5	44. 5	45. 7
46. 4	47. 5	48. 2	49. 8	50. 19
51. 3.75 gm	52. 3	53. 79.92 gm	54. 7	